

Q1 2021 (01 Sep Shift 2)

A capacitor is connected to a 20 V battery through a resistance of 10Ω . It is found that the potential difference across the capacitor rises to 2 V in $1\mu\text{s}$. The capacitance of the capacitor is μF . Given : $\ln\left(\frac{10}{9}\right) = 0.105$

(1) 9.52

(2) 0.95

(3) 0.105

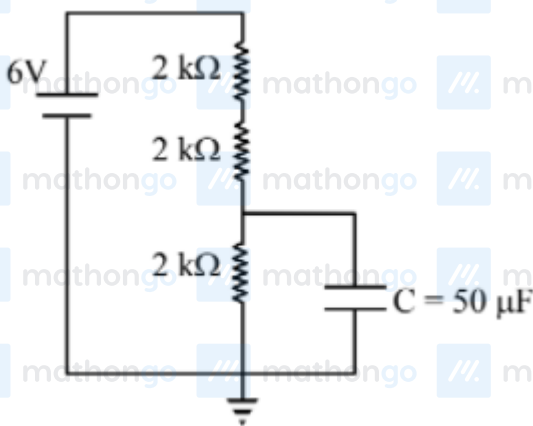
(4) 1.85

Q2 2021 (31 Aug Shift 2)

A parallel plate capacitor of capacitance $200\mu\text{F}$ is connected to a battery of 200 V. A dielectric slab of dielectric constant 2 is now inserted into the space between plates of capacitor while the battery remain connected. The change in the electrostatic energy in the capacitor will beJ.

Q3 2021 (31 Aug Shift 1)

A capacitor of $50\mu\text{F}$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is μC .



Q4 2021 (27 Aug Shift 2)

Three capacitors $C_1 = 2\mu\text{F}$, $C_2 = 6\mu\text{F}$ and $C_3 = 12\mu\text{F}$ are connected as shown in figure. Find the ratio of the charges on capacitors C_1 , C_2 and C_3 respectively:

A

(1) $2 : 1 : 1$

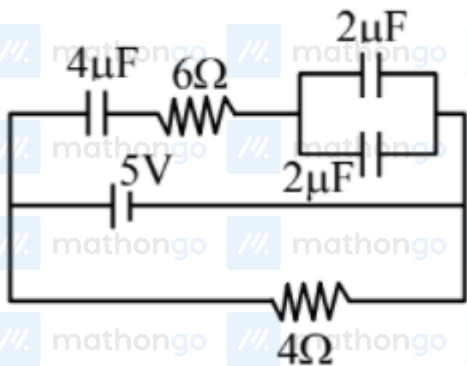
(2) $2 : 3 : 3$

(3) $1 : 2 : 2$

(4) $3 : 4 : 4$

Q5 2021 (27 Aug Shift 1)

Calculate the amount of charge on capacitor of $4\mu\text{F}$. The internal resistance of battery is 1Ω :



(1) $8\mu\text{C}$

(2) zero

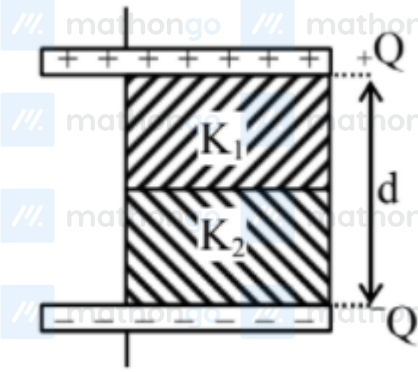
(3) $16\mu\text{C}$

(4) $4\mu\text{C}$

Q6 2021 (26 Aug Shift 2)

A parallel - plate capacitor with plate area A has separation d between the plates. Two dielectric slabs of dielectric constant K_1 and K_2 of same area $A/2$ and thickness $d/2$ are inserted in the space between the plates.

The capacitance of the capacitor will be given by:



- (1) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{K_1 + K_2} \right)$
- (2) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{2(K_1 + K_2)} \right)$
- (3) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{K_1 + K_2}{K_1 K_2} \right)$
- (4) $\frac{\epsilon_0 A}{d} \left(\frac{1}{2} + \frac{2(K_1 + K_2)}{K_1 K_2} \right)$

Q7 2021 (26 Aug Shift 1)

The material filled between the plates of a parallel plate capacitor has resistivity $200\Omega\text{m}$. The value of capacitance of the capacitor is 2pF . If a potential difference of 40 V is applied across the plates of the capacitor, then the value of leakage current flowing out of the capacitor is: (given the value of relative permittivity of material is 50)

- (1) $9.0\mu\text{A}$
- (2) 9.0 mA
- (3) 0.9 mA
- (4) $0.9\mu\text{A}$

Answer Key

Q1 (2)	Q2 (4)	Q3 (100)	Q4 (3)
Q5 (1)	Q6 (1)	Q7 (3)	

Q1 (2)

$$V = V_0 (1 - e^{-t/RC})$$

$$2 = 20 (1 - e^{-1/RC})$$

$$\frac{1}{10} = 1 - e^{-t/RC}$$

$$e^{-t/RC} = \frac{9}{10}$$

$$e^{t/RC} = \frac{10}{9}$$

$$\frac{t}{RC} = \ln\left(\frac{10}{9}\right) \Rightarrow C = \frac{t}{R \ln\left(\frac{10}{9}\right)}$$

$$C = \frac{10^{-6}}{10 \times 1.05} = .95 \mu\text{F}$$

Option (2)

Q2 (4)

$$\Delta U = \frac{1}{2}(\Delta C)V^2$$

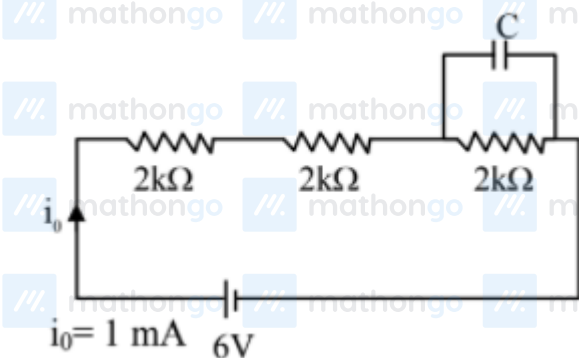
$$\Delta U = \frac{1}{2}(KC - C)V^2$$

$$\Delta U = \frac{1}{2}(2 - 1)CV^2$$

$$\Delta U = \frac{1}{2} \times 200 \times 10^{-6} \times 200 \times 200$$

$$\Delta U = 4J$$

Q3 (100)

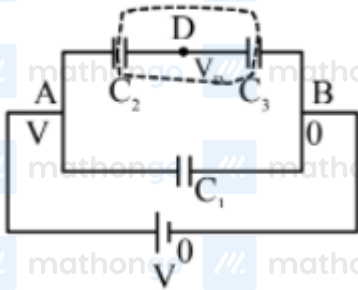


Pot. Diff. across each resistor = 2 V

$$q = CV$$

$$= 50 \times 10^{-6} \times 2 = 100 \times 10^{-6} = 100 \mu\text{C}$$

Q4 (3)



$$(V_D - V) C_2 + (V_D - 0) C_3 = 0$$

$$(V_D - V) 6 + (V_D - 0) 12 = 0$$

$$V_D - V + 2 V_D = 0$$

$$V_D = \frac{V}{3}$$

$$q_2 = (V - V_D) C_2 = \left(V - \frac{V}{3}\right) (6 \mu\text{F})$$

$$q_2 = (4 V) \mu\text{F}$$

$$q_3 = (V_D - 0) C_3 = \frac{V}{3} \times 12 \mu\text{F} = 4 V \mu\text{F}$$

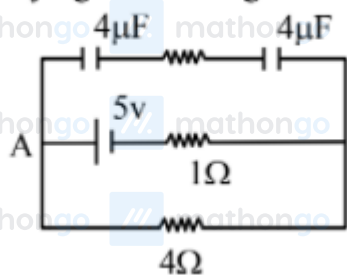
$$q_1 = (V - 0) C_1 = V(2 \mu\text{F})$$

$$q_1 : q_2 : q_3 = 2 : 4 : 4$$

$$q_1 : q_2 : q_3 = 1 : 2 : 2$$

Q5 (1)

On simplifying circuit we get



No current in upper wire.

$$\therefore V_{AB} = \frac{5}{4+1} \times 4 = 4\text{V}$$

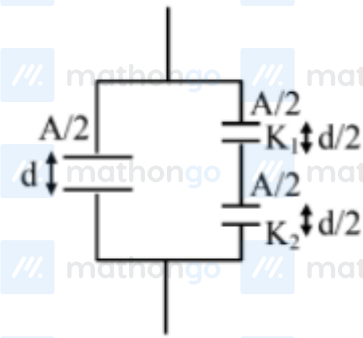
$$\therefore \theta = (C_{\text{eq}}) V$$

$$\Rightarrow 2 \times 4 = 8\mu\text{C}$$

Q6 (1)

$$C_{\text{eq}} = \frac{\frac{A}{2}\epsilon_0}{d} + \frac{A\epsilon_0}{d} \frac{K_1 K_2}{K_1 + K_2}$$

$$= \frac{A\epsilon_0}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{K_1 + K_2} \right)$$



Q7 (3)

$$\rho = 200\Omega\text{m}$$

$$C = 2 \times 10^{-12} \text{ F}$$

$$V = 40 \text{ V}$$

$$K = 56$$

$$i = \frac{q}{\rho k \epsilon_0} = \frac{q_0}{\rho k \epsilon_0} e^{-\frac{t}{\rho k \epsilon_0}}$$

$$i_{\text{max}} = \frac{2 \times 10^{-12} \times 40}{200 \times 50 \times 8.85 \times 10^{-12}}$$

$$= \frac{80}{10^4 \times 8.85} = 903\mu\text{A} = 0.9 \text{ mA}$$

Option (3)